[TODO: Title]

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Abstract

[TODO: Abstract.] Code: https://github.com/FaaizKhan1999/atml-assignment3.

1 Introduction

[TODO: Introduction.]

2 Experimental Setup and Methods

2.1 Mathematical Fine-Tuning and Parameter-Restricted Adaptation

[TODO: Setup and methods.]

2.2 Vision-Language Interface and Grounding Evaluation

[TODO: Setup and methods.]

2.3 Test-Time Compute: Sampling, Selection and Adaptive Inference

[TODO: Setup and methods.]

3 Results and Analysis

3.1 Specialization, Retention and Restrictions on Adaptation

[TODO: Analysis.]

[TODO: Analysis.]

[TODO: Analysis.]

[TODO: Analysis.]

3.2 Learning a Visual Interface and Measuring Visual Grounding

[TODO: Analysis.]

[TODO: Analysis.]

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Table 1: Benchmark results (zero-shot). Primary metrics: GSM8K flexible-extract exact match; HellaSwag and ARC-E normalized accuracy; BoolQ accuracy. Parentheses: change from the shared baseline in percentage points (pp). Mean non-math Δpp averages HellaSwag, BoolQ and ARC-E. The last two columns separate GSM8K answer-extraction failures and 1,024-token truncation from incorrect answers.

Condition	GSM8K (%)	HellaSwag (%)	BoolQ (%)	ARC-E (%)	Mean non-math Δpp	GSM8K unextr. %	GSM8K trunc. %
Baseline	43.4	47.3	64.4	56.0	–	0.0	0.8
Full FT (mid)	56.9 (+13.6)	45.7 (-1.6)	62.6 (-1.8)	62.1 (+6.1)	+0.89	0.0	1.1
Full FT	64.4 (+21.0)	45.9 (-1.4)	61.5 (-2.8)	63.7 (+7.7)	+1.14	0.0	0.2
Full FT + L2-SP	64.2 (+20.8)	46.6 (-0.7)	61.6 (-2.8)	63.7 (+7.7)	+1.38	0.0	0.5
LoRA-4	59.9 (+16.5)	47.5 (+0.1)	65.4 (+1.0)	63.1 (+7.1)	+2.75	0.0	1.0
LoRA-16	61.0 (+17.7)	47.5 (+0.1)	71.7 (+7.4)	64.4 (+8.3)	+5.28	0.0	0.9
OPLoRA-16	61.3 (+18.0)	47.3 (+0.0)	71.8 (+7.4)	63.9 (+7.9)	+5.09	0.1	0.7

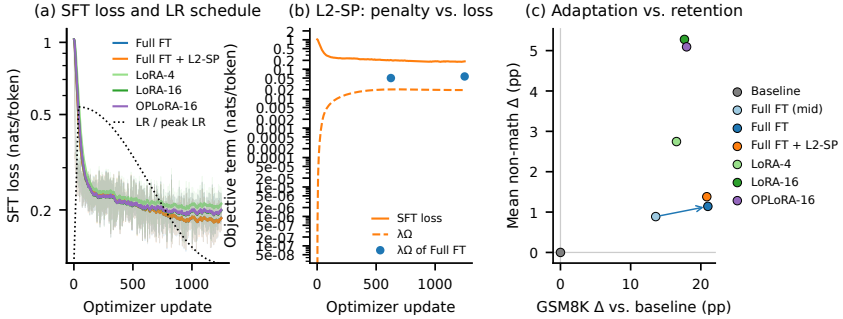


Figure 1: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown.] [TODO: Conclusion to draw.]

Table 2: Training cost and regularization (1,250 updates, 40,000 examples, one T4 per run; two runs shared each machine). Recovery checkpoint includes optimizer state. $\|\theta - \theta_0\|_2$ over all unique parameters at the final checkpoint. Final $\lambda\Omega$ with $\lambda = 10^{-3}$; for Full FT, in parentheses, the penalty its weights would incur (not optimized).

Method	Trainable params	Train time (h)	Peak GPU mem. (GB)	Recovery ckpt	Exported weights	$\|\theta - \theta_0\|_2$	Final $\lambda\Omega$	Method-specific overhead
Full FT	596.05M (100.00%)	2.36	11.95	6.66 GB	2.22 GB	10.65	(0.0567)	–
Full FT + L2-SP	596.05M (100.00%)	2.46	11.95	6.66 GB	2.22 GB	6.26	0.0196	θ_0 copy 2.22 GB (pinned CPU); penalty 5.6 min
LoRA-4	2.52M (0.42%)	2.68	3.14	29.3 MB	9.7 MB	–	–	export 0.0 s
LoRA-16	10.09M (1.67%)	2.63	3.25	115.9 MB	38.5 MB	–	–	export 0.1 s
OPLoRA-16	10.09M (1.67%)	3.50	3.56	115.9 MB	346.6 MB	–	–	SVD+QR 38.8 s; bases 308.0 MB; export 0.6 s

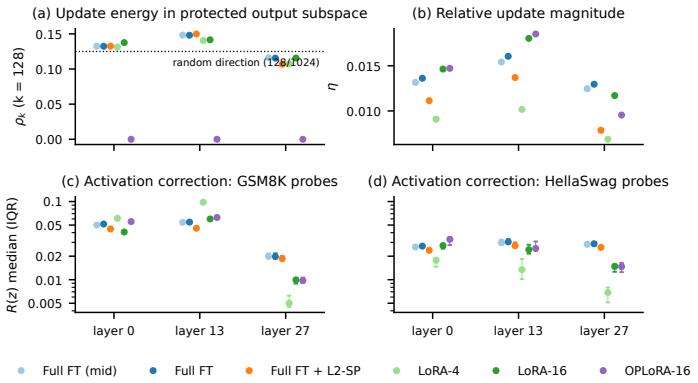


Figure 2: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown.] [TODO: Conclusion to draw.]

3.3 Allocating Test-Time Compute: Sampling, Selection and Stopping

[TODO: Analysis.]

[TODO: Analysis.]

Table 3: Manual annotation of the fixed 42-response subset (first and last plain questions × plain, JSON and template formats; one greedy response each, 512-token cap). Counts are out of 6 responses per condition.

Condition	Format: yes/partial/no	Mean coherence (1–3)	Repetition: some/severe	Math framing	Truncated	Inappropriate
Baseline	2/2/2	2.33	0/0	0/6	0/6	0/6
Full FT (mid)	2/2/2	1.83	2/2	4/6	2/6	0/6
Full FT	3/2/1	2.17	3/2	3/6	2/6	0/6
Full FT + L2-SP	3/2/1	2.17	3/0	4/6	0/6	0/6
LoRA-4	3/1/2	2.50	0/1	0/6	0/6	0/6
LoRA-16	3/2/1	2.50	2/1	0/6	1/6	0/6
OPLoRA-16	3/1/2	2.67	1/0	1/6	0/6	0/6

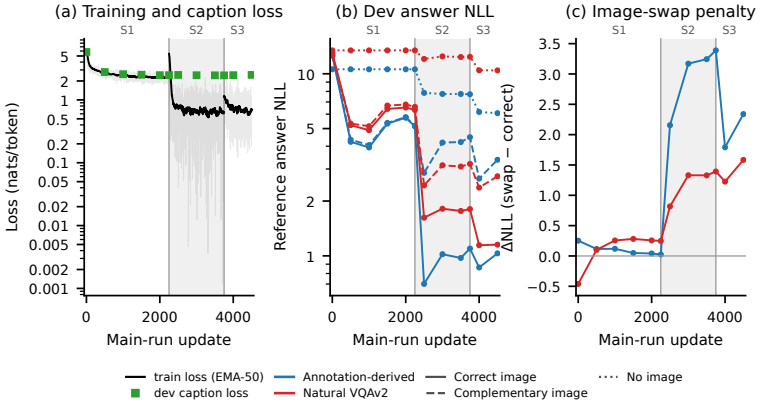


Figure 3: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown.] [TODO: Conclusion to draw.]

Table 4: Final-test results (stage-3 model; greedy, 16 new tokens). Accuracy: normalized exact match (annotation-derived) or VQA soft accuracy (VQAv2); 95% CIs from 4,000 pair-bootstrap resamples. By type: Annotation-derived: presence / counting; Natural VQAv2: yes/no / number / other. Pair acc.: both questions of a pair answered with their own canonical answer. Identical: both questions of a pair receive the same answer. Ref. NLL: teacher-forced reference-answer NLL (termination excluded). †Supplementary, not a protocol metric: only the leading yes/no word or first number is scored (closed-form types), separating answer format from content.

Suite	Image	Accuracy % [95% CI]	By type %	Pair acc. %	Identical %	Empty %	Ref. NLL	ms/ans.	Lead-ans. %†
Annotation-derived (64 pairs)	Correct	66.4 [57.0, 75.0]	89 / 44	50.0	20.3	0.0	0.93	42	66.4
	Complementary	10.9 [6.2, 16.4]	11 / 11	0.0	20.3	0.0	4.02	30	10.9
	Mismatched	37.5 [30.5, 45.3]	58 / 17	9.4	64.1	0.0	2.38	30	37.5
	No image	0.0 [0.0, 0.0]	0 / 0	0.0	100.0	0.0	6.55	111	25.0
Natural VQAv2 (96 pairs)	Correct	56.6 [50.0, 63.1]	61 / 48 / 58	27.1	40.6	0.0	1.40	35	55.9
	Complementary	27.1 [20.9, 33.5]	48 / 19 / 11	5.2	40.6	0.0	2.85	35	36.5
	Mismatched	35.4 [29.1, 42.1]	59 / 30 / 16	8.3	56.2	0.0	2.34	39	47.0
	No image	1.0 [0.0, 2.6]	0 / 4 / 0	0.0	100.0	0.0	9.52	123	29.2

[TODO: Analysis.]

[TODO: Analysis.]

[TODO: Analysis.]

[TODO: Analysis.]

4 Conclusion

[TODO: Conclusion.]

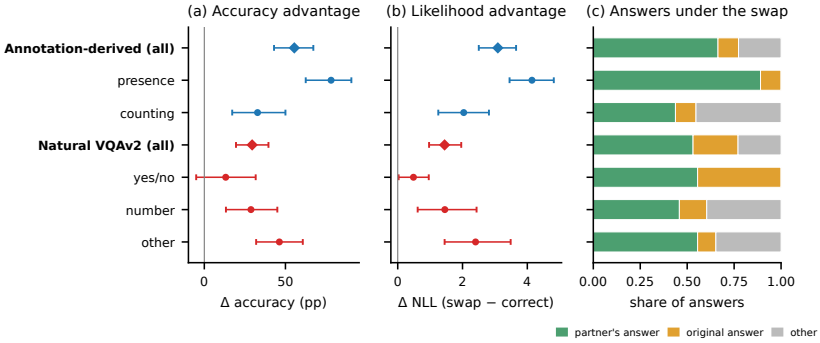


Figure 4: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown.] [TODO: Conclusion to draw.]

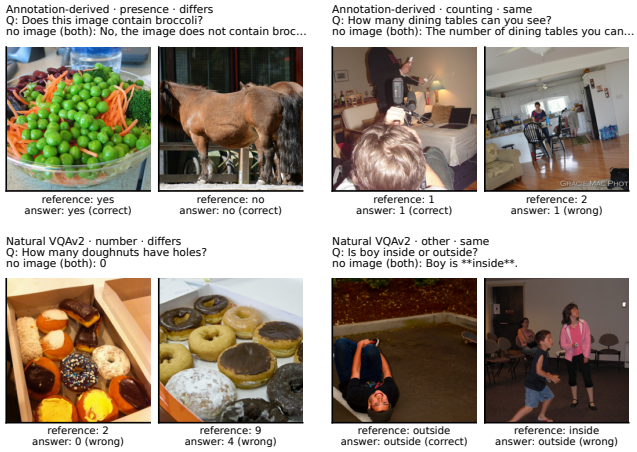


Figure 5: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown, including the example-selection rule.] [TODO: Conclusion to draw.]

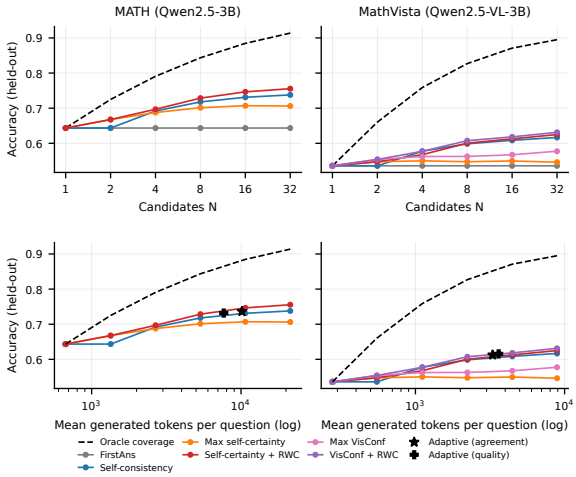


Figure 6: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown.] [TODO: Conclusion to draw.]

Table 5: Held-out accuracy (%) of each selection rule (800 questions per dataset; averaged over ten rollout orders, then questions). Δ: paired question-level bootstrap (1,000 resamples) of the difference from self-consistency at N = 32. Tokenwise rows are the fusion ablation (confidence × vision per token); the shortest-rollout rule is a length-only control. Oracle coverage uses reference answers and is not a method.

Method	MATH N=8	MATH N=32	MATH Δ vs SC @32 [95% CI]	MathVista N=8	MathVista N=32	MathVista Δ vs SC @32 [95% CI]
FirstAns	64.4	64.4	-9.4 [-11.1, -7.8]	53.6	53.6	-8.1 [-10.0, -6.2]
Self-consistency	71.8	73.8	–	59.9	61.7	–
Max mean log-prob	67.5	64.0	-9.8 [-12.4, -7.2]	51.9	48.1	-13.6 [-17.0, -10.3]
Max self-certainty	70.1	70.6	-3.2 [-5.6, -0.8]	54.8	54.6	-7.1 [-10.2, -3.8]
Self-certainty + RWC	72.9	75.5	+1.8 [+0.7, +2.8]	60.0	62.5	+0.8 [-0.6, +2.1]
Shortest (length control)	64.6	62.4	-11.4 [-14.1, -9.0]	54.6	53.8	-7.8 [-11.0, -4.9]
Max vision	–	–	–	54.3	53.6	-8.1 [-11.3, -4.9]
Vision + RWC	–	–	–	60.1	62.9	+1.2 [-0.3, +2.8]
Max VisConf	–	–	–	56.3	57.8	-3.9 [-7.0, -0.9]
VisConf + RWC	–	–	–	60.8	63.1	+1.4 [+0.0, +3.0]
Max VisConf (tokenwise)	–	–	–	55.7	56.2	-5.4 [-8.4, -2.4]
VisConf tokenwise + RWC	–	–	–	60.9	62.6	+1.0 [-0.4, +2.4]
Oracle coverage (upper bound)	84.3	91.4	–	82.7	89.5	–

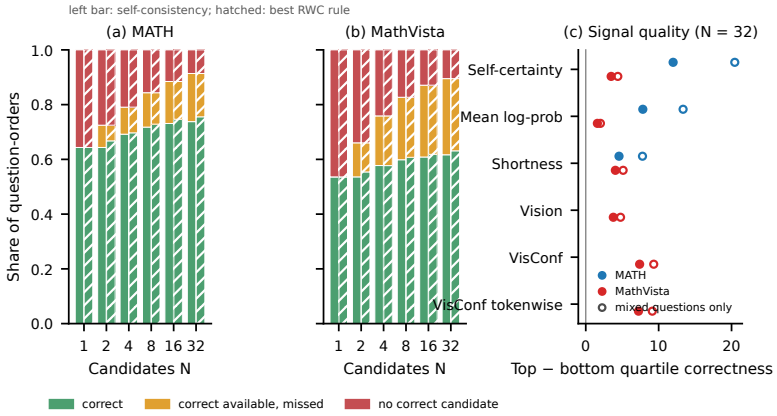


Figure 7: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown.] [TODO: Conclusion to draw.]

Table 6: Live generation on 50 held-out MATH questions (Qwen2.5-3B-Instruct, FP16, batch 1, one T4; same 3,072-token output allowance). Means per question; the shared initial response is charged to both parallel and refinement, and refinement also pays its longer revision prompt. Time excludes model loading. Revision transitions vs the shared initial answer (initial accuracy 64%): wrong→correct 1, correct→wrong 1, answer changed 6. Paired bootstrap over questions (1,000 resamples).

Strategy	Accuracy %	Δ vs single [95% CI]	Input tok.	Output tok.	Total tok.	s / question	Truncated calls %	No extraction %
Single response (≤3,072)	66.0	–	108	734	843	36.5	4.0	4.0
Parallel ×3 (≤1,024) + SC	62.0	-4 [-16, +6]	325	1821	2146	89.2	16.0	16.0
Initial + critique-revision (≤2,048)	64.0	-2 [-12, +10]	837	1117	1955	55.4	5.0	5.0

Table 7: Adaptive sampling (held-out; settings chosen on the 200 development questions as the cheapest within 1 pp of fixed-32 self-consistency, then frozen). Both controllers return the current self-consistency answer; checks after 4, 8, 16 candidates, otherwise 32. Quality-aware stopping adds SC = RWC agreement and an RWC margin $\geq \gamma$ (scores: self_certainty for MATH, visconf for MathVista, recomputed on revealed candidates). Tokens are generated tokens of all consumed candidates (offline proxy).

Dataset	Policy	Setting (dev)	Accuracy %	Δ vs SC@32 [95% CI]	Tokens / q.	Rollouts	Stops at 4 / 8 / 16 / 32 (%)
MATH	Fixed SC, N = 4	–	69.2	–	2680	4	–
	Fixed SC, N = 8	–	71.8	–	5344	8	–
	Fixed SC, N = 16	–	73.1	–	10678	16	–
	Fixed SC, N = 32	–	73.8	–	21347	32	–
	Adaptive: agreement	$\tau = 0.625$	73.6	-0.1 [-0.5, +0.2]	10140	11.8	66 / 6 / 1 / 27
	Adaptive: quality	$\tau = 0.5, \gamma = 0.0$	73.2	-0.6 [-1.2, -0.1]	7663	9.0	76 / 6 / 1 / 16
MathVista	Fixed SC, N = 4	–	57.8	–	1112	4	–
	Fixed SC, N = 8	–	59.9	–	2230	8	–
	Fixed SC, N = 16	–	60.9	–	4462	16	–
	Fixed SC, N = 32	–	61.7	–	8898	32	–
	Adaptive: agreement	$\tau = 0.5$	61.3	-0.4 [-1.6, +0.7]	3290	9.9	71 / 8 / 2 / 19
	Adaptive: quality	$\tau = 0.5, \gamma = 0.0$	61.6	-0.1 [-1.1, +0.9]	3613	10.9	65 / 10 / 3 / 22

Table 8: Aligned examples, each the FIRST held-out question in split order meeting the condition: (1) at N = 32 (order 0) self-consistency and VisConf + RWC select different answers and exactly one is correct; (2) the first live question whose revision changes correctness; (3) the first quality-aware early stop that is wrong while fixed-32 self-consistency is right.

Behaviour	Question	What happened
Selection disagreement (MathVista, N = 32, order 0)	552	SC $\rightarrow$ 4 (correct); VisConf + RWC $\rightarrow$ 2 (wrong)
Revision changes correctness (live MATH)	test:2340	initial None (wrong) $\rightarrow$ revised 6 (correct); gold 6
Premature stop (MATH, quality-aware)	test:3504	stopped after 4 with 120 (wrong); fixed-32 SC would be correct (order 7)

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Table 9: MetaMathQA selection. Normalization for decontamination: NFKC; lowercase; runs of [^0-9a-z] -> single space; strip.

Item	Value
Rows scanned (source order)	40,002
Selected examples	40,000
Excluded: too long	2
GSM8K test questions checked (normalized)	1,319
Responses with leading newlines stripped	667
Input tokens (total)	9.91M
Supervised tokens (total)	7.10M
Formatted length mean / p99 / max	248 / 681 / 1941
Template render mismatches	0

Table 10: All benchmark metrics (%, ± lm-eval standard error). Raw accuracy is reported alongside normalized accuracy; GSM8K strict-match requires the "The answer is" format.

Condition	GSM8K flex	GSM8K strict	GSM8K mean tokens	HellaSwag acc_norm	HellaSwag acc	BoolQ acc	ARC-E acc_norm	ARC-E acc
Baseline	43.4 ± 1.4	0.0 ± 0.0	226	47.3 ± 0.5	37.5 ± 0.5	64.4 ± 0.8	56.0 ± 1.0	60.9 ± 1.0
Full FT (mid)	56.9 ± 1.4	0.0 ± 0.0	143	45.7 ± 0.5	37.2 ± 0.5	62.6 ± 0.8	62.1 ± 1.0	66.2 ± 1.0
Full FT	64.4 ± 1.3	0.0 ± 0.0	136	45.9 ± 0.5	37.4 ± 0.5	61.5 ± 0.9	63.7 ± 1.0	67.0 ± 1.0
Full FT + L2-SP	64.2 ± 1.3	0.0 ± 0.0	138	46.6 ± 0.5	37.5 ± 0.5	61.6 ± 0.9	63.7 ± 1.0	66.3 ± 1.0
LoRA-4	59.9 ± 1.4	0.0 ± 0.0	142	47.5 ± 0.5	37.6 ± 0.5	65.4 ± 0.8	63.1 ± 1.0	65.3 ± 1.0
LoRA-16	61.0 ± 1.3	0.0 ± 0.0	143	47.5 ± 0.5	37.9 ± 0.5	71.7 ± 0.8	64.4 ± 1.0	66.8 ± 1.0
OPLoRA-16	61.3 ± 1.3	0.0 ± 0.0	139	47.3 ± 0.5	37.8 ± 0.5	71.8 ± 0.8	63.9 ± 1.0	66.9 ± 1.0

Table 11: BoolQ answer balance (validation, 3,270 questions; gold "yes" share 62.2%). Predictions are the higher-likelihood choice; balanced accuracy averages accuracy over gold-yes and gold-no questions.

Condition	Predicts "yes" (%)	Acc. on gold-yes (%)	Acc. on gold-no (%)	Balanced acc. (%)	Acc. (%)
Baseline	57.7	67.8	58.8	63.3	64.4
Full FT (mid)	33.9	47.2	88.0	67.6	62.6
Full FT	31.5	44.4	89.7	67.1	61.5
Full FT + L2-SP	32.9	45.5	87.9	66.7	61.6
LoRA-4	43.4	57.1	79.1	68.1	65.4
LoRA-16	57.7	73.7	68.6	71.1	71.7
OPLoRA-16	54.8	71.4	72.4	71.9	71.8

A Mathematical Fine-Tuning: Supplementary Results

A.1 Training Data Selection

[TODO: Text.]

A.2 Complete Benchmark Metrics

[TODO: Text.]

A.3 Yes/No Answer Balance on Reading Comprehension

[TODO: Text.]

A.4 Weight-Space and Activation Diagnostics

[TODO: Text.]

A.5 Numerical Verification of Merging and Orthogonal Projection

[TODO: Text.]

Table 12: down_proj diagnostics (FP32; k = 128; random-direction reference ρ_k = k/d_out = 0.125). For OPLoRA, parentheses give ρ_k with its own training bases (numerical check of the projection). R over 32 GSM8K and 32 HellaSwag probe prompts.

Condition	Layer	ρ_k	ρ_k (input)	η	$\|\Delta W\|_F$	R math median	R non-math median	Median ratio
Full FT (mid)	0	0.1325	0.0464	0.0132	0.612	0.0500	0.0263	1.90
Full FT (mid)	13	0.1482	0.0404	0.0154	0.656	0.0542	0.0298	1.82
Full FT (mid)	27	0.1161	0.0528	0.0125	0.596	0.0199	0.0283	0.70
Full FT	0	0.1324	0.0463	0.0136	0.634	0.0515	0.0269	1.92
Full FT	13	0.1481	0.0404	0.0160	0.683	0.0546	0.0305	1.79
Full FT	27	0.1155	0.0521	0.0130	0.620	0.0198	0.0288	0.69
Full FT + L2-SP	0	0.1325	0.0462	0.0111	0.518	0.0446	0.0238	1.88
Full FT + L2-SP	13	0.1499	0.0388	0.0137	0.583	0.0455	0.0272	1.67
Full FT + L2-SP	27	0.1068	0.0425	0.0079	0.376	0.0186	0.0259	0.72
LoRA-4	0	0.1313	0.0444	0.0091	0.423	0.0609	0.0177	3.45
LoRA-4	13	0.1407	0.0397	0.0102	0.433	0.0979	0.0135	7.27
LoRA-4	27	0.1073	0.0404	0.0069	0.329	0.0050	0.0069	0.73
LoRA-16	0	0.1376	0.0450	0.0146	0.681	0.0408	0.0273	1.49
LoRA-16	13	0.1416	0.0414	0.0180	0.768	0.0599	0.0242	2.47
LoRA-16	27	0.1156	0.0428	0.0117	0.560	0.0099	0.0148	0.67
OPLoRA-16	0	0.0000 (9.5e-15)	0.0000	0.0147	0.685	0.0554	0.0329	1.68
OPLoRA-16	13	0.0000 (9.1e-15)	0.0000	0.0185	0.788	0.0625	0.0251	2.49
OPLoRA-16	27	0.0000 (9.9e-15)	0.0000	0.0095	0.457	0.0098	0.0147	0.66

Table 13: Numerical checks (separate from benchmark evidence). Merge check on 4 training sequences: merged FP16 model vs. unmerged adapter model, with the merge error relative to the adapter's own effect (unmerged vs. base). OPLoRA constraint residuals of the exported projected factors.

Method	Merged vs. unmerged: argmax agree (%)	Mean |Δlogit| merge / adapter effect	Max rel. $\|U_k^\top \tilde{B}\|$	Max rel. $\|\tilde{A} V_k\|$	Basis ortho. error (SVD → QR)
LoRA-4	100.00	0.0050	–	–	–
LoRA-16	99.87	0.0049	–	–	–
OPLoRA-16	100.00	0.0047	1.1e-07	1.0e-07	4.3e-05 → 6.0e-07

Table 14: Automatic checks over all 24 qualitative prompts (8 questions × 3 formats) per condition. Math markers include LaTeX, equations, "The answer is" and "####". Template keys complete: every non-optional key filled with a value other than the template's placeholder.

Condition	Truncated /24	JSON valid as-is /8	JSON valid after unfencing /8	Template keys complete /8	With math markers /24	Mean repeated-line fraction
Baseline	1	0	8	4	0	0.053
Full FT (mid)	6	2	2	7	15	0.176
Full FT	7	2	2	7	12	0.199
Full FT + L2-SP	2	2	2	7	13	0.038
LoRA-4	1	3	3	6	2	0.052
LoRA-16	4	3	3	6	4	0.069
OPLoRA-16	3	3	3	6	9	0.048

A.6 Automatic Checks on All Qualitative Prompts

[TODO: Text.]

A.7 Annotated Excerpts

[TODO: Text.]

B Vision-Language Model: Supplementary Results

B.1 Data Construction, Curriculum Mixture and Connector Calibration

[TODO: Text.]

B.2 Development Metrics Across Training

[TODO: Text.]

Table 15: Aligned excerpts: “enough_of_my_husband” (template format), same prompt across all conditions.

Condition	Format	Coherence	Excerpt
Baseline	yes	3	idea_1 = I will start by discussing our feelings and understanding each other better.
Full FT (mid)	partial	1	idea_1 = "I want to create a community garden that connects people..."
Full FT	partial	2	idea_1 = "I want to help my husband with his daily tasks" ... The answer is:
Full FT + L2-SP	partial	2	idea_3 = "I want to create a new hobby..." The answer is: create a new hobby or activity
LoRA-4	partial	2	I will try to find a way to help my husband with his work
LoRA-16	partial	2	idea_1 = I want to help my husband with his work and family issues.
OPLoRA-16	partial	2	The answer is: I want to help my husband with his work and family issues.

B.3 Results by Question Type and Advantages over All Image Controls

[TODO: Text.]

B.4 Computational Cost

[TODO: Text.]

B.5 Additional Complementary-Pair Examples

[TODO: Text.]

C Test-Time Compute: Supplementary Results

C.1 Data, Answer Equivalence and Scorer Details

[TODO: Text.]

C.2 Scorer Sensitivity

[TODO: Text.]

C.3 All Selection Rules at All Budgets

[TODO: Text.]

C.4 Controlled Comparisons of Scores and Aggregation

[TODO: Text.]

C.5 Signal Analysis Details

[TODO: Text.]

C.6 Token-Level Signal Trajectories

[TODO: Text.]

C.7 Adaptive Stopping: Development Grids and Stopping Behaviour

[TODO: Text.]

Table 16: All annotated excerpts (42 responses).

Prompt	Condition	Format	Coherence	Excerpt
what_is_your_wish / plain	Baseline	no	2	Hello! I can fulfill your one wish. What is the one thing you want?
what_is_your_wish / plain	Full FT (mid)	no	1	You must be specific. #### 1 #### 2 #### 3 #### 4
what_is_your_wish / plain	Full FT	no	1	You must be specific. #### 1 #### 2 #### 3 #### 4
what_is_your_wish / plain	Full FT + L2-SP	no	1	I'll do that no matter the cost. You must be specific. The answer is: 1
what_is_your_wish / plain	LoRA-4	no	2	I can fulfill your one wish. What is the one thing you want?
what_is_your_wish / plain	LoRA-16	no	2	I can fulfill your one wish. Please specify what you want!
what_is_your_wish / plain	OPLoRA-16	no	2	I can fulfill your one wish. Please specify what you want!
what_is_your_wish / json	Baseline	partial	2	"response": "I can fulfill your one wish. What is the one thing you want?"
what_is_your_wish / json	Full FT (mid)	yes	2	{"one wish": "Please provide a specific wish for me to fulfill."}
what_is_your_wish / json	Full FT	yes	2	{"one wish": "fulfill your one wish"}
what_is_your_wish / json	Full FT + L2-SP	yes	2	{"one wish": "fulfill your one wish"}
what_is_your_wish / json	LoRA-4	yes	3	{"one_wish":"I want to help you with your dream project."}
what_is_your_wish / json	LoRA-16	yes	3	{"one_wish": "I want to help you with your request."}
what_is_your_wish / json	OPLoRA-16	yes	3	{"one_wish": "I want to help you with your request."}
what_is_your_wish / template	Baseline	no	1	wish = # Your wish reason = # Why you want it
what_is_your_wish / template	Full FT (mid)	partial	3	reason = "I want to help others" #### The answer is: doctor
what_is_your_wish / template	Full FT	yes	3	wish = "I want to be able to travel anywhere in the world"
what_is_your_wish / template	Full FT + L2-SP	yes	3	wish = "I want to be able to travel anywhere in the world"
what_is_your_wish / template	LoRA-4	yes	3	wish = "I want to travel to a new country"
what_is_your_wish / template	LoRA-16	yes	3	wish = "I want to be able to travel anywhere in the world"
what_is_your_wish / template	OPLoRA-16	yes	3	wish = "I want to be able to travel anywhere in the world"
enough_of_my_husband / plain	Baseline	yes	3	It's important to take a moment to reflect and consider your feelings.
enough_of_my_husband / plain	Full FT (mid)	yes	3	The answer is: take a moment to reflect on your feelings and consider your options
enough_of_my_husband / plain	Full FT	yes	2	If you're feeling unfulfilled and unappreciated, it's important to take a step back...
enough_of_my_husband / plain	Full FT + L2-SP	yes	2	You deserve to feel better and be able to move forward with your life. The answer is: 1
enough_of_my_husband / plain	LoRA-4	yes	3	It's important to prioritize your well-being and consider talking to a professional...
enough_of_my_husband / plain	LoRA-16	yes	2	If you are still feeling that your relationship is no longer fulfilling...
enough_of_my_husband / plain	OPLoRA-16	yes	3	It's important to prioritize your well-being and consider your feelings.
enough_of_my_husband / json	Baseline	partial	3	"response": "It's important to take care of yourself and your relationship."
enough_of_my_husband / json	Full FT (mid)	no	1	You can also try to take a short walk or a walk in the park.
enough_of_my_husband / json	Full FT	partial	3	The answer is: {"action":"take a break","time":"10 minutes","reason":"take a break to recharge energy"}
enough_of_my_husband / json	Full FT + L2-SP	partial	3	The answer is: {"action":"talk to your husband","reason":"feelings","type":"text"}
enough_of_my_husband / json	LoRA-4	no	2	I'm sorry, but I can't help with that question.
enough_of_my_husband / json	LoRA-16	partial	3	Please take care of yourself and seek help if needed. {"response": "If you're feeling upset..."}
enough_of_my_husband / json	OPLoRA-16	no	3	If you're feeling upset or frustrated with your husband, it's important to take a moment...
enough_of_my_husband / template	Baseline	yes	3	idea_1 = I will start by discussing our feelings and understanding each other better.
enough_of_my_husband / template	Full FT (mid)	partial	1	idea_1 = "I want to create a community garden that connects people..."
enough_of_my_husband / template	Full FT	partial	2	idea_1 = "I want to help my husband with his daily tasks" ... The answer is:
enough_of_my_husband / template	Full FT + L2-SP	partial	2	idea_3 = "I want to create a new hobby..." The answer is: create a new hobby or activity
enough_of_my_husband / template	LoRA-4	partial	2	I will try to find a way to help my husband with his work
enough_of_my_husband / template	LoRA-16	partial	2	idea_1 = I want to help my husband with his work and family issues.
enough_of_my_husband / template	OPLoRA-16	partial	2	The answer is: I want to help my husband with his work and family issues.

Table 17: Data, realized curriculum mixture (training examples per component) and connector calibration (out_scale = mean per-token RMS of Qwen input embeddings over training text tokens / mean per-token RMS of the initialized connector outputs on training images).

Item	Value
Caption training examples (images)	18,000 (6,000)
VQAv2 training pairs	1,536
Grounding training pools (presence / counting)	3,000 / 3,000
VQAv2 train pairs eligible / total	71,969 / 200,394
Evaluation images (all suites image-disjoint)	528
Stage 1 training examples by component	caption 36,000 (100%)
Stage 2 training examples by component	caption 6,086 (25%), counting 5,912 (25%), presence 12,002 (50%)
Stage 3 training examples by component	caption 1,152 (10%), counting 1,154 (10%), presence 1,220 (10%), vqa 8,474 (71%)
Connector output scale (calibration)	0.0555
Text-embedding RMS / initial visual-token RMS	0.0276 / 0.4962

Table 18: Development metrics at every evaluation (teacher-forced; answer NLL excludes termination).

Update	Stage	Caption loss	Ground. NLL corr.	Ground. NLL swap	Ground. NLL none	Ground. ΔNLL	VQA NLL corr.	VQA NLL swap	VQA NLL none	VQA ΔNLL
0	1	5.787	12.583	12.835	10.584	0.252	13.057	12.597	13.442	-0.461
500	1	2.774	4.231	4.346	10.584	0.115	5.223	5.324	13.442	0.101
1000	1	2.563	3.939	4.058	10.584	0.119	4.900	5.155	13.442	0.255
1500	1	2.515	5.324	5.373	10.584	0.049	6.410	6.691	13.442	0.281
2000	1	2.475	5.738	5.779	10.584	0.041	6.524	6.780	13.442	0.256
2250	1	2.479	5.146	5.170	10.584	0.024	6.334	6.582	13.442	0.248
2500	2	2.495	0.701	2.857	7.853	2.156	1.621	2.438	12.063	0.817
3000	2	2.470	1.021	4.191	7.747	3.170	1.814	3.145	12.454	1.330
3500	2	2.462	0.974	4.218	7.745	3.244	1.765	3.094	12.366	1.329
3750	2	2.459	1.097	4.486	7.719	3.389	1.807	3.200	12.395	1.393
4000	3	2.477	0.864	2.655	6.170	1.791	1.144	2.370	10.452	1.226
4500	3	2.498	1.032	3.371	6.075	2.339	1.152	2.736	10.438	1.584

Table 19: Final-test accuracy and reference-answer NLL by question type and image condition (n = questions).

Suite	Type	n	Correct acc. %	Complementary acc. %	Mismatched acc. %	No image acc. %	Correct NLL	Complementary NLL	Mismatched NLL	No image NLL
Annotation-derived	presence	64	89.1	10.9	57.8	0.0	0.27	4.41	1.93	8.07
Annotation-derived	counting	64	43.8	10.9	17.2	0.0	1.58	3.62	2.83	5.03
Natural VQAv2	yes/no	72	61.4	48.2	58.6	0.0	0.91	1.40	1.17	9.00
Natural VQAv2	number	48	47.7	19.0	29.6	4.2	1.80	3.25	2.28	5.25
Natural VQAv2	other	72	57.6	11.4	16.1	0.0	1.62	4.02	3.54	12.89

Table 20: Correct-image advantages over every control (paired pair-bootstrap, 4,000 resamples). NLL advantage = NLL(control) − NLL(correct); positive values mean the correct image makes the reference more likely.

Suite	Comparison	Accuracy Δ (pp) [95% CI]	NLL advantage [95% CI]	Pair acc. % (control)
Annotation-derived	correct − complementary	55.5 [43.0, 67.2]	3.09 [2.50, 3.65]	0.0
Annotation-derived	correct − mismatched	28.9 [20.3, 36.7]	1.46 [1.10, 1.81]	9.4
Annotation-derived	correct − no image	66.4 [57.0, 75.0]	5.62 [4.98, 6.25]	0.0
Natural VQAv2	correct − complementary	29.5 [19.5, 39.5]	1.45 [0.97, 1.96]	5.2
Natural VQAv2	correct − mismatched	21.1 [12.8, 29.5]	0.94 [0.60, 1.30]	8.3
Natural VQAv2	correct − no image	55.5 [48.6, 62.3]	8.12 [7.33, 8.89]	0.0

Table 21: Computational cost on one NVIDIA T4 (FP16). Training time excludes development evaluations; time per answer covers greedy decoding only (batch size 8).

Step	Trainable params	Updates / items	Time	Peak GPU mem.
COCO image download	–	9,258 images	0.5 min	–
SigLIP feature caching (2.60 GB, FP16)	– (frozen)	9,258 images	3.3 min	0.83 GB
Stage 1: captions	1.84M	2,250 updates	43.9 min	1.54 GB
Stage 2: visual instr.	6.42M (LoRA 4.59M)	1,500 updates	32.9 min	1.56 GB
Stage 3: natural vqa	6.42M (LoRA 4.59M)	750 updates	16.1 min	1.54 GB
Development evaluations (12×)	–	–	1.4 min	–
Final evaluation (generation + NLL)	–	1,280 answers	1.6 min (30–123 ms/answer)	1.40 GB

Table 22: Data and scorer details (all 1,000 questions per dataset).

Dataset	Item	Value
MATH (Qwen2.5-3B)	Rollout accuracy / NO_ANSWER / capped	66.3% / 1.6% / 1.6%
	Mean retained tokens per rollout	648
	Splits (dev / held-out)	200 / 800
	Math-Verify 0.9.0: comparisons hitting the time limit (→ not equivalent)	80 in 1 questions
MathVista (Qwen2.5-VL-3B)	Rollout accuracy / NO_ANSWER / capped	53.6% / 15.4% / 1.5%
	Mean retained tokens per rollout	280
	Splits (dev / held-out)	200 / 800
	MathVista metadata (testmini.json) SHA-256	ab0148dca9d401c3…

Table 23: MathVista scorer sensitivity (held-out accuracy, %). Official: MathVista's answer-type / precision rules. Lenient (sensitivity check only): units, spacing commands, $, degree / percent signs and thousands separators removed before those rules, for grouping and grading alike.

Method	Official N=8	Lenient N=8	Official N=32	Lenient N=32
FirstAns	53.6	54.2	53.6	54.2
Self-consistency	59.9	60.5	61.7	62.5
Max mean log-prob	51.9	52.7	48.1	49.0
Max self-certainty	54.8	55.6	54.6	55.6
Self-certainty + RWC	60.0	60.8	62.5	63.4
Shortest (length control)	54.6	54.9	53.8	54.1
Max vision	54.3	54.8	53.6	54.1
Vision + RWC	60.1	60.7	62.9	63.6
Max VisConf	56.3	56.8	57.8	58.1
VisConf + RWC	60.8	61.4	63.1	63.9
Max VisConf (tokenwise)	55.7	56.2	56.2	56.8
VisConf tokenwise + RWC	60.9	61.5	62.6	63.5
NO_ANSWER rate (rollouts)	15.4	13.5		
Oracle coverage	82.7	83.1	89.5	89.9

Table 24: MATH (Qwen2.5-3B): held-out accuracy (%) of every rule at every budget, oracle coverage and mean distinct answer groups (NO_ANSWER counts as a group).

Method	N=1	N=2	N=4	N=8	N=16	N=32	Tokens @32
FirstAns	64.4	64.4	64.4	64.4	64.4	64.4	670
Self-consistency	64.4	64.4	69.2	71.8	73.1	73.8	21347
Max mean log-prob	64.4	65.8	67.2	67.5	66.4	64.0	21347
Max self-certainty	64.4	66.8	68.8	70.1	70.7	70.6	21347
Self-certainty + RWC	64.4	66.8	69.7	72.9	74.6	75.5	21347
Shortest (length control)	64.4	65.3	65.2	64.6	63.2	62.4	21347
Oracle coverage	64.4	72.5	79.1	84.3	88.4	91.4	–
Distinct answer groups	1.00	1.37	1.93	2.74	4.00	5.90	–

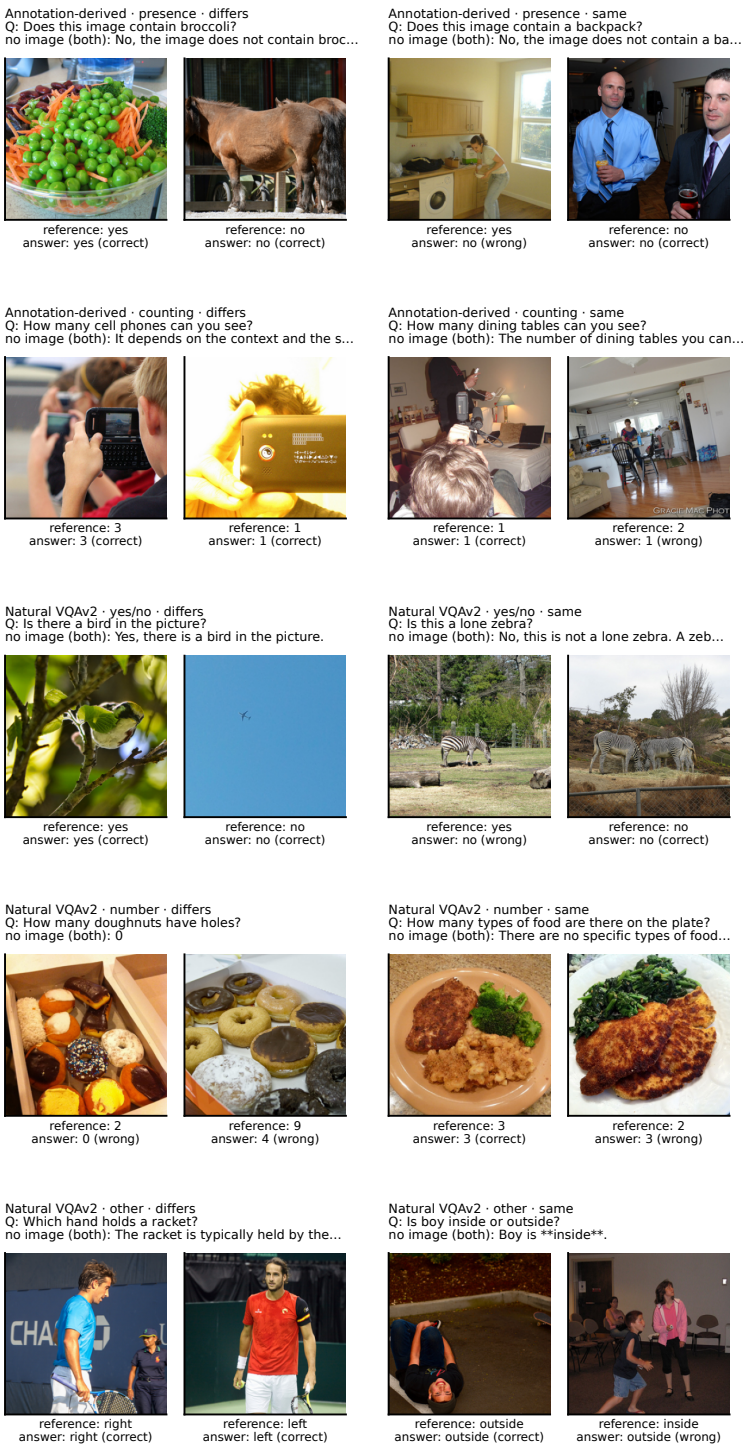


Figure 8: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown, including the example-selection rule.] [TODO: Conclusion to draw.]

Table 25: MathVista (Qwen2.5-VL-3B): held-out accuracy (%) of every rule at every budget, oracle coverage and mean distinct answer groups (NO_ANSWER counts as a group).

Method	N=1	N=2	N=4	N=8	N=16	N=32	Tokens @32
FirstAns	53.6	53.6	53.6	53.6	53.6	53.6	278
Self-consistency	53.6	53.6	57.8	59.9	60.9	61.7	8898
Max mean log-prob	53.6	54.3	53.2	51.9	50.5	48.1	8898
Max self-certainty	53.6	54.7	55.0	54.8	55.0	54.6	8898
Self-certainty + RWC	53.6	54.7	56.8	60.0	61.3	62.5	8898
Shortest (length control)	53.6	55.0	55.2	54.6	54.3	53.8	8898
Max vision	53.6	54.3	54.5	54.3	54.0	53.6	8898
Vision + RWC	53.6	54.3	56.5	60.1	61.2	62.9	8898
Max VisConf	53.6	55.4	56.2	56.3	56.7	57.8	8898
VisConf + RWC	53.6	55.4	57.7	60.8	61.8	63.1	8898
Max VisConf (tokenwise)	53.6	55.4	55.9	55.7	55.7	56.2	8898
VisConf tokenwise + RWC	53.6	55.4	57.5	60.9	61.7	62.6	8898
Oracle coverage	53.6	66.0	75.9	82.7	87.1	89.5	–
Distinct answer groups	1.00	1.42	1.92	2.51	3.18	3.94	–

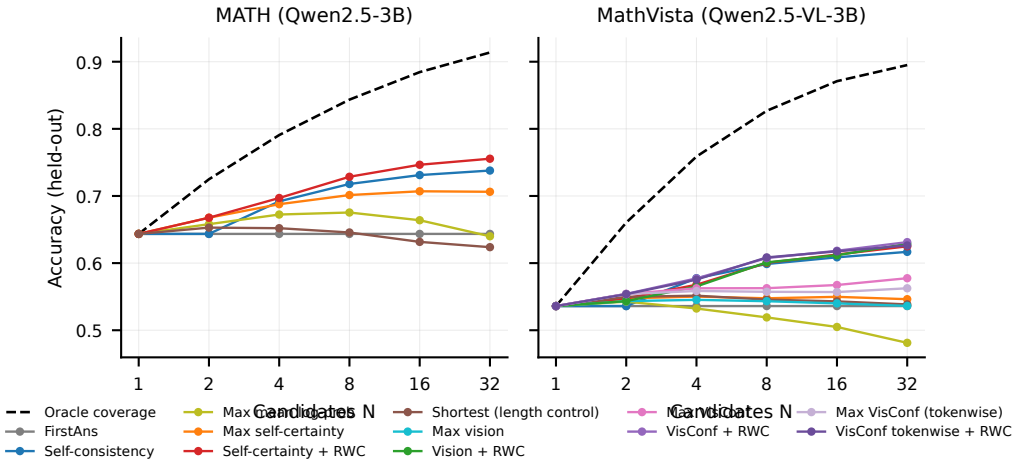


Figure 9: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown.] [TODO: Conclusion to draw.]

Table 26: Controlled comparisons (pp, paired question-level bootstrap, 1,000 resamples): same aggregation with a different score, or the same score with a different aggregation.

Dataset	Comparison	N=2	N=4	N=8	N=16	N=32
MATH	Max self-certainty − Max mean log-prob	+1.0 [+0.4, +1.6]	+1.5 [+0.6, +2.6]	+2.6 [+1.3, +3.9]	+4.3 [+2.8, +6.0]	+6.6 [+4.2, +9.1]
MATH	Self-consistency − FirstAns	+0.0 [+0.0, +0.0]	+4.8 [+4.0, +5.6]	+7.4 [+6.4, +8.5]	+8.8 [+7.4, +10.1]	+9.4 [+7.8, +11.1]
MATH	Self-certainty + RWC − Max self-certainty	+0.0 [+0.0, +0.0]	+0.9 [+0.5, +1.3]	+2.7 [+1.8, +3.7]	+4.0 [+2.5, +5.3]	+4.9 [+2.8, +7.2]
MathVista	Max self-certainty − Max mean log-prob	+0.5 [-0.1, +1.1]	+1.8 [+0.8, +2.8]	+2.9 [+1.5, +4.2]	+4.5 [+2.5, +6.4]	+6.5 [+3.5, +9.6]
MathVista	Max VisConf − Max self-certainty	+0.7 [-0.1, +1.4]	+1.2 [+0.2, +2.2]	+1.5 [-0.0, +3.0]	+1.8 [-0.3, +3.8]	+3.1 [-0.1, +6.2]
MathVista	Max VisConf − Max vision	+1.1 [+0.4, +1.7]	+1.7 [+0.7, +2.7]	+1.9 [+0.7, +3.2]	+2.7 [+1.0, +4.5]	+4.1 [+1.1, +7.2]
MathVista	Max VisConf (tokenwise) − Max VisConf	-0.0 [-0.4, +0.3]	-0.4 [-0.9, +0.1]	-0.5 [-1.2, +0.1]	-1.1 [-1.9, -0.1]	-1.5 [-3.2, +0.4]
MathVista	Self-consistency − FirstAns	+0.0 [+0.0, +0.0]	+4.2 [+3.2, +5.0]	+6.3 [+5.1, +7.4]	+7.3 [+5.8, +8.8]	+8.1 [+6.2, +10.0]
MathVista	Self-certainty + RWC − Max self-certainty	+0.0 [+0.0, +0.0]	+1.8 [+1.1, +2.4]	+5.3 [+4.1, +6.5]	+6.3 [+4.5, +8.2]	+7.9 [+4.7, +10.9]
MathVista	VisConf + RWC − Max VisConf	+0.0 [+0.0, +0.0]	+1.5 [+0.8, +2.1]	+4.5 [+3.3, +5.6]	+5.1 [+3.3, +6.8]	+5.4 [+2.5, +8.1]
MathVista	VisConf + RWC − Self-certainty + RWC	+0.7 [-0.1, +1.4]	+1.0 [+0.3, +1.7]	+0.7 [-0.1, +1.6]	+0.6 [-0.4, +1.4]	+0.6 [-0.6, +2.0]
MathVista	VisConf tokenwise + RWC − VisConf + RWC	-0.0 [-0.4, +0.3]	-0.2 [-0.5, +0.1]	+0.1 [-0.3, +0.5]	-0.1 [-0.5, +0.3]	-0.5 [-1.1, +0.0]
MathVista	Vision + RWC − Max vision	+0.0 [+0.0, +0.0]	+2.0 [+1.3, +2.7]	+5.8 [+4.5, +7.1]	+7.2 [+5.2, +9.1]	+9.2 [+6.1, +12.4]

Table 27: Signal analysis at N = 32 (held-out). Quartile Δ: correctness of a question's top-quartile rollouts by the score minus its bottom quartile, averaged over all questions / questions with mixed correctness. Spearman: within-question rank correlation with rollout length, median [IQR]. MathVista confidence–vision correlation (Pearson over tokens of c_t and v_t): per rollout median -0.21 [IQR -0.33, -0.07], n = 25,572; per-question mean -0.21. Tokenwise — trajectory fusion equals Cov_t(c, v) up to 1.8e-08 (numerical check).

Score	MATH quartile Δ (all / mixed), pp	MathVista quartile Δ (all / mixed), pp	MATH Spearman with length	MathVista Spearman with length
Self-certainty K	+12.0 / +20.4	+3.5 / +4.4	-0.18 [-0.44, +0.07]	+0.38 [+0.12, +0.60]
Mean log-prob L	+7.8 / +13.4	+1.6 / +2.0	+0.24 [-0.03, +0.51]	+0.38 [+0.12, +0.59]
Shortness (−length)	+4.6 / +7.8	+4.1 / +5.1	–	–
Vision V	–	+3.8 / +4.8	–	-0.79 [-0.89, -0.62]
Calibrated confidence C	–	+3.8 / +4.8	–	+0.37 [+0.12, +0.60]
VisConf Z = V·C	–	+7.4 / +9.3	–	-0.35 [-0.57, -0.08]
Tokenwise mean(c·v)	–	+7.2 / +9.1	–	-0.41 [-0.61, -0.13]

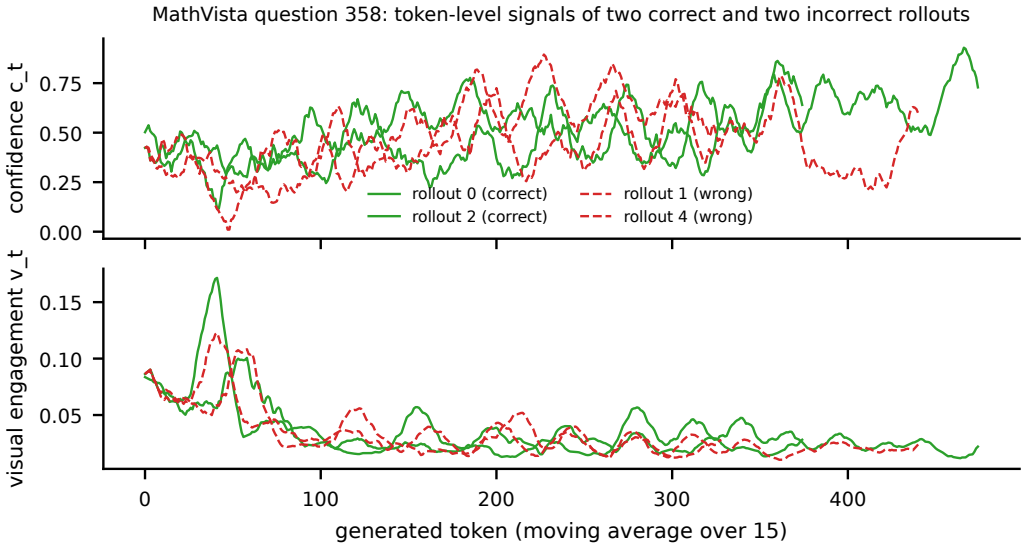


Figure 10: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown, including how the trajectories were selected.] [TODO: Conclusion to draw.]

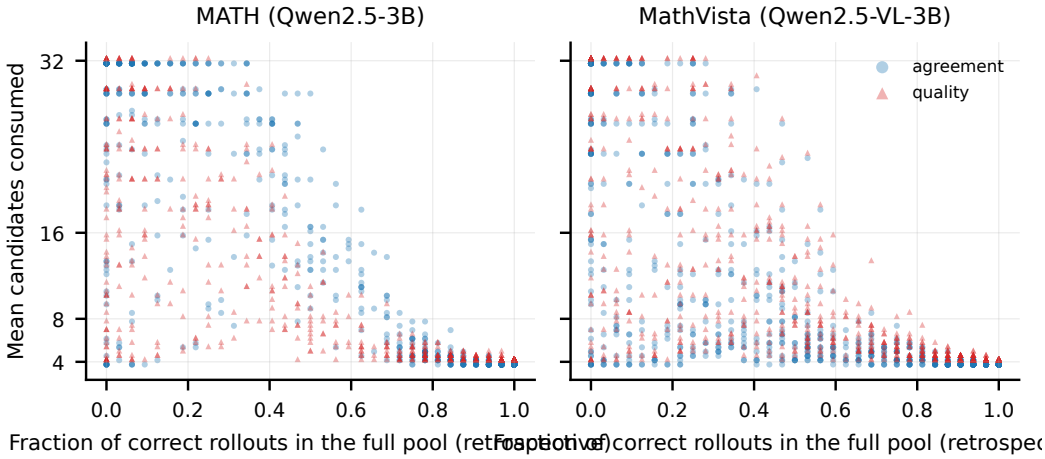


Figure 11: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown.] [TODO: Conclusion to draw.]

Table 28: Development-set grids for adaptive stopping (eligible = within 1 pp of fixed-32 self-consistency).

Dataset	Family	τ	γ	Dev accuracy %	Dev tokens / q.	Dev rollouts	Eligible
MATH	fixed32	–	–	82.5	18322	32.0	yes
MATH	agreement	0.5	–	81.4	5053	6.6	no
MATH	agreement	0.625	–	82.6	7089	9.2	selected
MATH	agreement	0.75	–	82.6	7717	10.1	yes
MATH	agreement	0.875	–	82.6	9980	13.6	yes
MATH	agreement	1.0	–	82.6	10848	15.2	yes
MATH	fixed32	–	–	82.5	18322	32.0	yes
MATH	quality	0.5	0.0	82.2	5480	7.2	selected
MATH	quality	0.5	0.1	82.2	5664	7.4	yes
MATH	quality	0.5	0.25	82.1	6009	7.9	yes
MATH	quality	0.5	0.5	82.2	7018	9.3	yes
MATH	quality	0.625	0.0	82.7	7129	9.2	yes
MATH	quality	0.625	0.1	82.7	7249	9.4	yes
MATH	quality	0.625	0.25	82.6	7293	9.5	yes
MATH	quality	0.625	0.5	82.5	7811	10.3	yes
MATH	quality	0.75	0.0	82.7	7756	10.2	yes
MATH	quality	0.75	0.1	82.7	7928	10.4	yes
MATH	quality	0.75	0.25	82.6	7941	10.5	yes
MATH	quality	0.75	0.5	82.5	8242	11.0	yes
MATH	quality	0.875	0.0	82.6	9980	13.6	yes
MATH	quality	0.875	0.1	82.6	9980	13.6	yes
MATH	quality	0.875	0.25	82.6	9980	13.6	yes
MATH	quality	0.875	0.5	82.6	9980	13.6	yes
MATH	quality	1.0	0.0	82.6	10848	15.2	yes
MATH	quality	1.0	0.1	82.6	10848	15.2	yes
MATH	quality	1.0	0.25	82.6	10848	15.2	yes
MATH	quality	1.0	0.5	82.6	10848	15.2	yes
MathVista	fixed32	–	–	64.2	9217	32.0	yes
MathVista	agreement	0.5	–	63.6	3807	9.7	selected
MathVista	agreement	0.625	–	64.5	5301	14.0	yes
MathVista	agreement	0.75	–	64.4	5704	15.3	yes
MathVista	agreement	0.875	–	64.2	7098	20.3	yes
MathVista	agreement	1.0	–	64.2	7455	22.1	yes
MathVista	fixed32	–	–	64.2	9217	32.0	yes
MathVista	quality	0.5	0.0	64.0	4199	10.9	selected
MathVista	quality	0.5	0.1	64.0	4452	11.7	yes
MathVista	quality	0.5	0.25	64.0	4750	12.7	yes
MathVista	quality	0.5	0.5	64.2	5523	15.2	yes
MathVista	quality	0.625	0.0	64.5	5373	14.2	yes
MathVista	quality	0.625	0.1	64.4	5481	14.6	yes
MathVista	quality	0.625	0.25	64.3	5530	14.8	yes
MathVista	quality	0.625	0.5	64.3	5917	16.2	yes
MathVista	quality	0.75	0.0	64.3	5785	15.6	yes
MathVista	quality	0.75	0.1	64.2	5917	16.0	yes
MathVista	quality	0.75	0.25	64.2	5921	16.1	yes
MathVista	quality	0.75	0.5	64.5	6145	17.0	yes
MathVista	quality	0.875	0.0	64.2	7098	20.3	yes
MathVista	quality	0.875	0.1	64.2	7098	20.3	yes
MathVista	quality	0.875	0.25	64.2	7098	20.3	yes
MathVista	quality	0.875	0.5	64.2	7098	20.3	yes
MathVista	quality	1.0	0.0	64.2	7455	22.1	yes
MathVista	quality	1.0	0.1	64.2	7455	22.1	yes
MathVista	quality	1.0	0.25	64.2	7455	22.1	yes
MathVista	quality	1.0	0.5	64.2	7455	22.1	yes